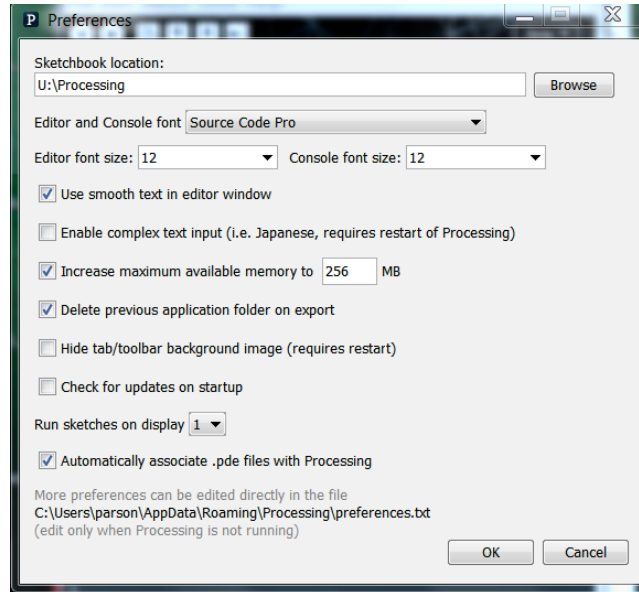


CSC 120 Introduction to Creative Graphical Coding, Spring 2021

Prof. Earl, Assignment 3, Implementing and testing a class that models your avatar.

This assignment is due via **D2L Assignment 3** by **11:59 PM on Friday April 16**. When using Processing on the Kutztown campus Windows computers, make sure to start out **every time** by setting your Processing Preferences -> Sketchbook Location to U:\Processing. The U:\ drive is a networked drive that will save your work and make it accessible across campus. If you save it to your desktop or the lab PC you are using, you will lose your work when you log out. You must save it to the U:\ drive. If you do not have a folder called Processing under U:\, you must create one using the Windows Explorer. Processing Preferences is under the File menu on Windows.



Use your working solution to assignment 2 as a starting point. If it has deficiencies, you must fix them. Please visit my office hours for assistance with this.

Purpose: Animate an avatar using Processing. Write a class & an array.

Steps in going from assignment 2 to assignment 3 example:

1. Create "class Avatar" around definition of function avatar(...).
Increase indent for the function. It is now inside the class.
Change the function's name from "avatar" to "display".
2. Remove function parameters x, y, scale, and any other parameters to this function from its parameter list, and make them into data fields (variables) inside the class.
3. Create another function "void move()" inside the Avatar class, and put all of the code to update the Avatar's location and other Avatar data fields into move(). Make sure to change the variable names in the move() code to match the Avatar data fields, e.g., avx, avy, etc. I needed to create two Avatar fields "avxspeed" and "axyspeed" to take the place of global variables avatarXspeed and avatarYspeed. Increase indent as needed.
4. Create a constructor function Avatar(...) inside the class, with

one parameter for every data field in the class. Name the parameters something slightly different from the fields. Use assignment statements to assign the values from the parameters into the fields. The class definition is now complete. Next, we must use it.

5. Remove all **global** variables associated with Avatar state.
6. Create two new global variables "avatar1" and "avatar2" of class Avatar.
7. In the setup() function, call the constructor for Avatar twice, once to assign an object reference into avatar1, and the second into avatar2. Use constructor arguments to give the starting location, speed, and other Avatar state for these objects. The constructor parameters show what arguments are needed for the constructor. I pulled some of my values from this code in assignment 2:

```
avatar(avatarX, height/2, 1.0, legX); // initial value of avatarX and legX were 0.  
avatar(2 *width/3, height/5, 0.5, 0); // second call, without movement
```
8. Inside the draw() function, replace calls to the avatar() function of assignment 2 (which drew the avatar) with calls to the display() and move() functions of objects avatar1 and avatar2. I replaced this code:

```
avatar(avatarX, height/2, 1.0, legX); // initial value of avatarX and legX were 0.  
avatar(2 *width/3, height/5, 0.5, 0); // second call, without movement
```

with this code:

```
// Display, then move the avatars:  
avatar1.display();  
avatar2.display();  
avatar1.move();  
avatar2.move();
```
9. Create a global array variable "crowd" that holds 2 or more Avatar objects. Construct two Avatar objects and place them into this array in setup(). You can use the same parameters or different parameters as used for avatar1 and avatar2. We will cover arrays the week of April 6.
10. Comment out the avatar1 & avatar2 display() and move() code of step 8 (do not delete it), and add a loop (for or while loop) to display() and move() all of the Avatar objects in array crowd in its place.
11. Update documentation comments at the top of the sketch, also for class Avatar and each of its fields and functions, and for any other code that has changed in this assignment 3 code.

LATE PENALTY IS 10% PER DAY AS USUAL. TURN IT IN VIA D2L ASSIGNMENT 3.

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Drop avatarClass.pde into D2L Assignment3 by 11:59 PM on Friday April 16.